

PARTH SARTHI

Data Engineer | Data Architecture, Cloud Platforms & Enterprise Analytics

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SUMMARY

Data Engineer and Solution Architect with 10 years of experience designing, building and operating enterprise data platforms for large-scale analytics. Architected Johnson & Johnson's iDiscover data platform and AnGen analytics layer, led migration of 1,000+ ETL jobs to an AWS/Airflow framework, and currently supports iDiscover modernisation from Airflow to Databricks. Strong in scalable ETL/ELT, Python, PySpark, SQL, Redshift, Databricks, data modelling, CI/CD, cloud security, data quality, MDM and stakeholder-led solution delivery across Pharmaceutical, Commercial and Supply Chain domains.

CORE SKILLS

Data Engineering: Data integration, ingestion, ETL/ELT, structured data, metadata-driven pipelines, API integration, batch orchestration, data reconciliation

Cloud & Platforms: AWS, Amazon Redshift, Databricks, Apache Airflow, EMR, S3, Lambda, Glue, CloudWatch, PostgreSQL, Talend, Control-M

Programming: Python, PySpark, SQL, PL/SQL, T-SQL, Shell Scripting

Architecture & Governance: Enterprise data warehousing, dimensional modelling, star/snowflake schemas, mapping rules, MDM, data quality, IAM, sensitive-data access controls, data lineage, technical standards

DevOps & Delivery: CI/CD, Git, JIRA, Agile/Scrum, production monitoring, automated testing, stakeholder management, technical leadership

EXPERIENCE

Solution Architect / Data Engineer

Dec 2016 – Present

Tata Consultancy Services (TCS) – Johnson & Johnson | Singapore / India

Primary architect for iDiscover and AnGen enterprise data platforms; onsite in Singapore since Apr 2022.

- **Enabled enterprise-scale analytics by architecting iDiscover from inception**, designing ingestion, warehouse and downstream data-delivery layers that support Pharmaceutical Innovative Medicine, Commercial and Supply Chain use cases.
- **Reduced development effort and improved pipeline scalability by building a metadata-driven ETL framework**, standardising reusable, restartable and horizontally scalable ingestion and transformation patterns.
- **Modernised a large ETL estate by leading migration of 1,000+ Talend jobs to an AWS-based Apache Airflow framework**, improving maintainability and establishing scalable orchestration patterns for production workloads.
- **Improved analytics performance and consistency by designing Redshift and PostgreSQL warehouse models**, applying dimensional, star and snowflake modelling patterns to transform source data into governed Fact and Dimension datasets.
- **Improved data reliability by implementing Python/PySpark transformations, validation and source-to-target reconciliation**, including automated SIT comparison and JIRA evidence capture through TCS SPyDr integration.
- **Strengthened platform security by replacing access-key-based access with IAM role-based authentication**, aligning cloud data access with AWS security best practices and reducing credential-management risk.
- **Improved release velocity and reduced deployment risk by establishing CI/CD and automated regression processes**, while using CloudWatch, Airflow and operational controls to support production monitoring and SLA adherence.
- **Reduced infrastructure cost by optimising Amazon EMR configurations against workload requirements**, improving compute utilisation without compromising processing needs.
- **Enabled commercial performance analytics across markets by architecting AnGen solutions integrating commercial and medical data**, supporting governed KPIs for J&J vs competitor product sales, sales performance and market performance across countries.
- **Built 10+ governed data marts as AnGen's semantic analytics layer for Tableau**, standardising business logic and metric definitions used in KPI cards and downstream reporting.
- **Enabled medical-field analytics by modelling Medical Science Liaison interactions with healthcare professionals**, including communication mode, HCP coverage and identification of key impact makers.
- **Protected business metric integrity by governing data quality, mapping logic and pipeline reliability across iDiscover and AnGen**, serving as the primary technical point of contact for architecture and delivery.
- **Led a delivery portfolio of approximately 60 people across multiple projects for the same enterprise client**, aligning technical delivery, shared platform architecture and competing project priorities.
- **Strengthened database CI/CD governance as administrator for Flyway repositories**, reviewing, approving and merging developer pull requests and enforcing controlled database-code promotion and engineering standards.

Current Platform Modernisation – Airflow to Databricks

- **Advancing platform scalability by supporting migration of iDiscover orchestration and processing**

workloads from Apache Airflow toward Databricks, adopting managed data-engineering capabilities and modern pipeline patterns.

Supply Chain – OTIF (On Time In Full) | 2019

- **Enabled accurate supply-chain delivery measurement by engineering OTIF pipelines**, calculating expected delivery dates using warehouse holidays, public holidays, transit time and delivery-calendar dependencies and producing analytics-ready datasets.

Master Data Management (MDM)

- **Improved cross-system data consistency by maintaining master-data mappings** for Product, Therapeutic Area, Disease Area and related reference domains used in enterprise integration and downstream analytics.

DATA GOVERNANCE & DELIVERY LEADERSHIP

- **Established reusable engineering and governance practices** spanning metadata-driven pipeline patterns, dimensional-modelling conventions, MDM mappings, IAM-based access, code review, CI/CD and source-to-target reconciliation.
- **Safeguarded consistent reporting across commercial and medical analytics** by governing semantic data marts, KPI definitions, mapping logic and analytics-ready datasets used by Tableau consumers.

TECHNICAL SKILLS

Cloud	AWS (Redshift, EMR, S3, Lambda, Glue, IAM, CloudWatch), Databricks
Data Engineering	Apache Airflow, Talend ETL, AWS Glue, Control-M, metadata-driven ETL, data ingestion, ETL/ELT, reconciliation
Databases	Amazon Redshift, PostgreSQL, Oracle, MS SQL Server, MySQL
Languages	Python, PySpark, SQL, PL/SQL, T-SQL, Shell Scripting
Data Architecture	Data warehousing, semantic data marts, dimensional modelling, star schema, snowflake schema, ER modelling, Inmon, Kimball, MDM
Engineering	CI/CD, Flyway repository administration, Git, pull-request governance, JIRA, Agile/Scrum, automated testing, production monitoring, IAM, cloud security

CERTIFICATIONS

Databricks Certified Data Engineer Associate • Databricks Accredited: Databricks Fundamentals • Claude Certified Architect Professional • Claude Certified Developer Foundations • Claude Certified Associate Foundations

EDUCATION

Bachelor of Engineering – Computer Science and Engineering S J B Institute of Technology (SJBIT), Visvesvaraya Technological University, Bangalore	Jul 2012 – Jun 2016
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